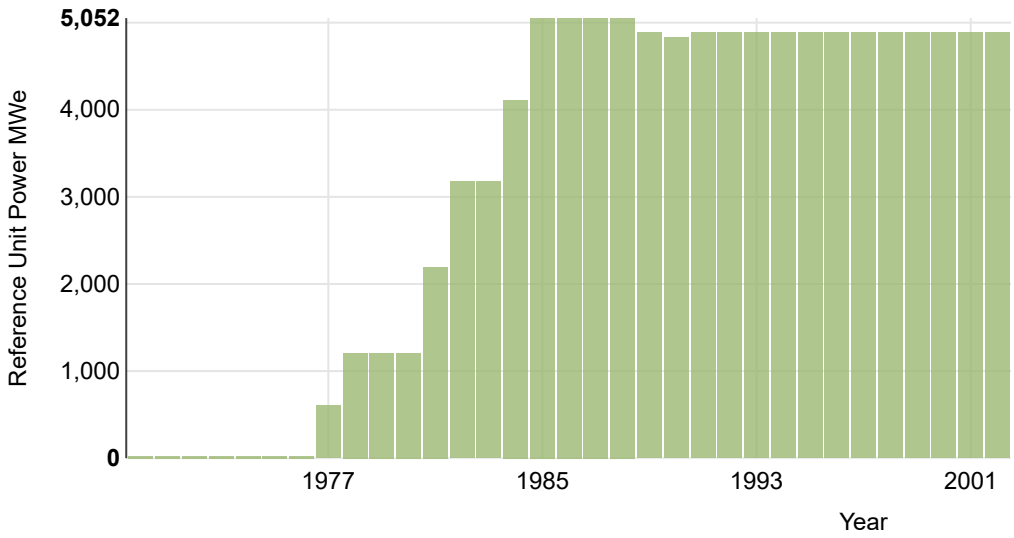
- Taiwan shut down its last nuclear reactor in May 2025.
- Two advanced reactors were under construction, but this project was cancelled.
- The Democratic Progressive Party elected in January 2016, and re-elected in 2020, introduced a policy of phasing out nuclear power by 2025.
- In December 2021, a majority of voters rejected the possibility of restarting construction of the two

Privacy - Terms

mothballed reactors at Lungmen.

0	0	6
OPERABLE REACTORS	REACTORS UNDER CONSTRUCTION	REACTORS SHUTDOWN
0 MWe	0 MWe	5,058 MWe

Operable nuclear power capacity

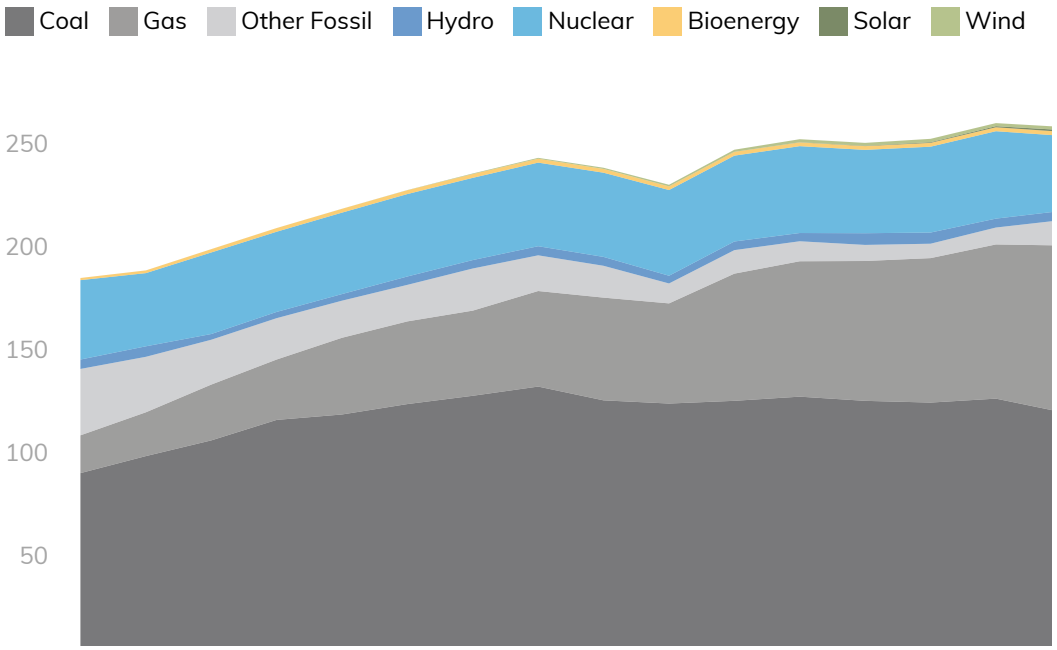


Electricity sector

Per capita electricity consumption in Taiwan has increased from about 8300 kWh in 2000 to about 11,800 kWh in 2023.

Electricity generation by source (TWh)

Taiwan electricity generation by source (TWh)



Source: EMBER Electricity Data Explorer

Generating capacity in 2024 was 57.9 GWe. Electricity consumption grew by 250% between 1990 and 2024. Nuclear power provided a significant part of the island's electricity supply for about four decades before the last reactor was shut down in May 2025.

Energy policy

Taiwan imports about 98% of its energy, which is vital to the rapidly industrializing economy. There has been a concerted programme to develop renewable capacity since the Renewable Energy Development Act of 2009. The government set ambitious plans for 20% of the island's electricity to be generated from renewable sources by 2025. In the event this target was missed by a wide margin.

Taiwan has now set a target of generating 60-70% of its electricity from renewables by 2050.

Nuclear phase-out policy

The Democratic Progressive Party elected in January 2016, and re-elected in 2020, had a policy of phasing out nuclear power by 2025. The MOEA had earlier said in April 2015 that the closure by 2025 of the three nuclear power plants that were then operating could result in:

- Lower economic growth rates.
- Higher levels of pollution.
- An increase of more than 10% in electricity prices.
- A 0.5% decline in Taiwan's GDP.
- Up to 15% increase in carbon dioxide emissions.

In November 2018 a specific referendum question on nuclear power in Taiwan showed apparent support (59%) for maintaining the island's significant nuclear power sector beyond 2025. A total of 5.89 million 'yes' votes and 4.01 million 'no' votes were cast in response to the question: "Do you agree that subparagraph 1, Article 95 of the Electricity Act, which reads: 'Nuclear-energy-based power-generating facilities shall wholly stop running by 2025,' should be abolished?" In May 2019 the Legislative Yuan removed the provision in Article 95 of the Electricity Act. However, the new energy strategy introduced in 2019 maintained the intention of the ruling political party to phase out nuclear power by 2025. The country's last reactor was shut down in May 2025.

Power outages

Power outages have been a consistent issue for Taiwan, with widespread power cuts in 2017, 2021 and 2022.

On 15 August 2017 – after running for several months with very low reserve margin, which fell below 2% a week earlier – a problem at a large gas-fired power plant resulted in a blackout across half of Taiwan for about five hours. World Nuclear Association said: "The Taiwanese government has allowed ideology to undermine public wellbeing by keeping nuclear capacity offline at a time when the country is struggling with power shortages." The National Association of Industry and Commerce called on the government to reconsider its reliance on natural gas and neglect of nuclear power, and to "entertain the possibility" of completing the 2700 MWe Lungmen nuclear plant, where the first unit had been almost completed (see below).

Nuclear power industry

Six reactors were connected to the grid at three sites on the island between 1977 and 1985. In 2019 it was announced that the

Chinshan plant would commence decommissioning, and in July 2021 Kuosheng 1 was shut down, followed by unit 2 in March 2023. Maanshan 1 was shut down in July 2024, followed by unit 2 in May 2025.

All reactors were initially expected to have 40-year operating lifetimes. In 2009 Taipower said that it planned to replace the steam generators of the two Maanshan pressurized water reactors (PWRs) by about 2020 if it could obtain operating licence extensions from the Atomic Energy Council (AEC). This and other work would have yielded uprates of some 440 MWe across the six reactors. In 2007 the AEC said that the Chinshan boiling water reactor (BWR) plant had undergone a safety evaluation and was safe to run for a further 20 years following the planned licence expiry in 2017. The AEC had approved this lifetime extension, though in November 2011 a new national energy policy disallowed it and affirmed simply a 40-year operating lifetime. Taipower had expected to seek 20-year licence renewals for all six reactors. In December 2013 Taipower submitted a new application for lifetime extension of Chinshan, and the AEC safety evaluation report was expected late in 2016. However, due to the new energy policy of the ruling Democratic Progressive Party, Taipower withdrew the application in July 2016, and AEC ceased its evaluation accordingly. In July 2019 the AEC approved decommissioning plans for both Chinshan units over a 25-year period.

Prior to the government's nuclear phaseout policy, Taipower had been in 2009 examining the prospects for six more reactors, starting with a pair at an established site to be online about 2020.

Lungmen

Two 1350 MWe Advanced Boiling Water Reactor (ABWR) units were under construction at Lungmen, near Taipei. Initial plans to procure the units on an engineering, procurement and construction (EPC) basis failed, and contracts were awarded to GE for the nuclear reactors, Mitsubishi for the turbines and others for the rest, making it a particularly difficult project to manage. Construction began in 1999 with intention of 2004 completion.

When the two reactors were one-third completed, a new cabinet cancelled the project but work resumed the following year later after legal appeal and a government resolution in favour. The project was thus significantly delayed by this, in addition to other delays. A date for completion of the first unit was to be announced early in 2012 – in June 2011 it was undergoing pre-operational

testing, with the second unit about a year behind. In January 2014 Taipower said unit 1 would come into operation in 2015 and unit 2 in 2017. "Full testing" of systems in unit 1 was then due to be completed in June 2014. However, in April 2014 in response to political discord the government said that unit 1 would be mothballed after safety checks had been completed and construction of unit 2 would be halted. In July 2015 the unit was sealed, meaning that equipment has been put into a protected condition which would allow future use.

As a result of cost escalation due to the construction hiatus plus project management and engineering problems, almost NT$ 300 billion ($9.9 billion) has been spent on the project. An AEC minister in March 2019 said that $1.9-2.3 billion would be required to complete the plant.

In February 2019 Taipower ruled out starting up the plant. It stated that it would take six to seven years to start commercial operation, and that GE would not be able to replace many of the ageing components installed 20 years ago as the company had ceased production of many of them.

A referendum took place in December 2021, in which a majority of voters rejected the possibility of restarting construction of the two mothballed reactors.

Taiwan shutdown nuclear reactors

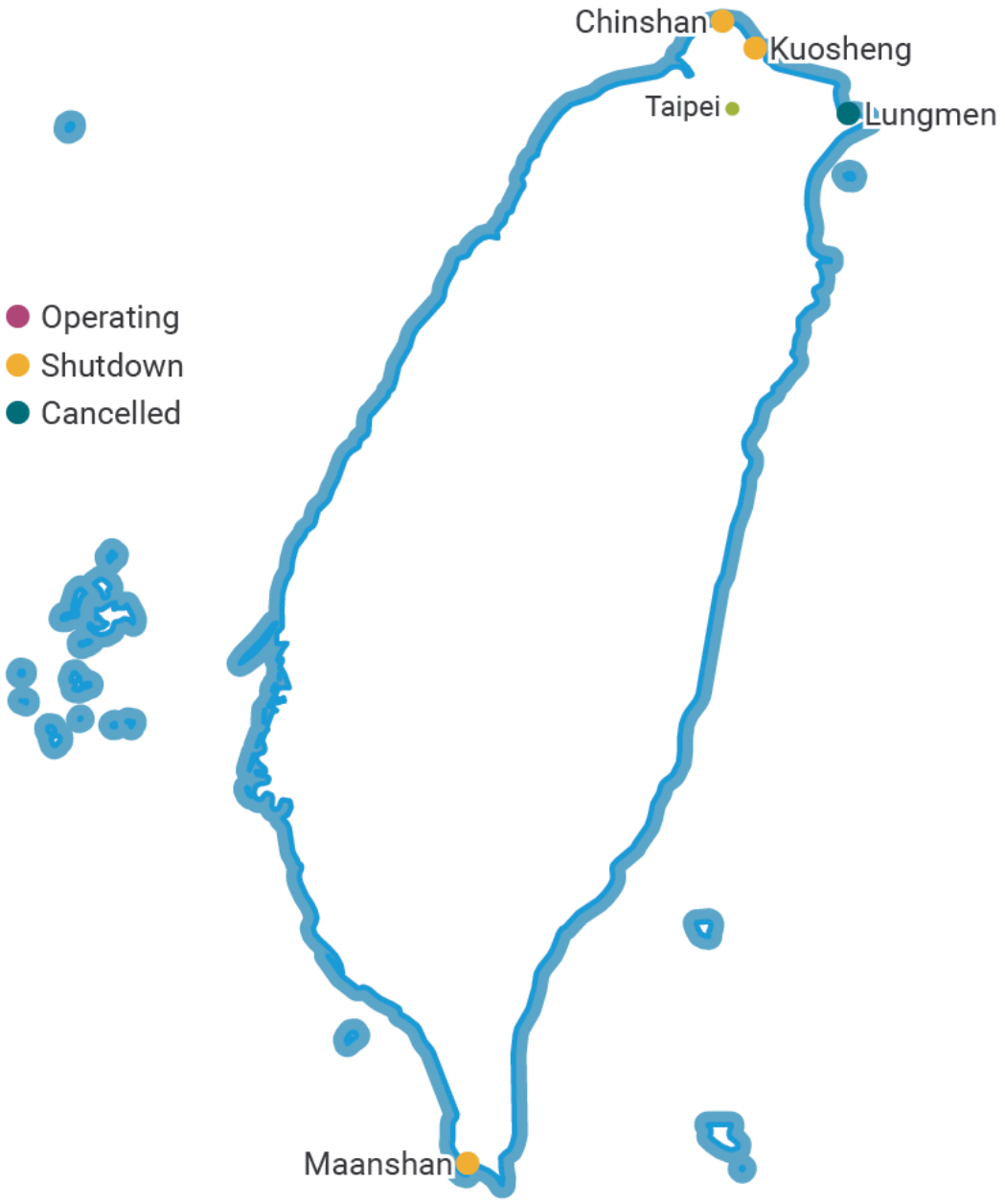
Reactor name	Model	Reactor type	Net capacity (MWe)	Permanent shutdown date
Chinshan 1	BWR-4	BWR	604	2018-12-06
Chinshan 2	BWR-4	BWR	604	2019-07-16
Kuosheng 1	BWR-6	BWR	985	2021-12-28
Kuosheng 2	BWR-6	BWR	985	2023-03-15
Maanshan 1	WE 312 (3 loop-12 foot)	PWR	936	2024-07-28
Maanshan 2	WE 312 (3 loop-12 foot)	PWR	938	2025-05-18

Taiwan nuclear reactors cancelled

	Type	MWe net
Lungmen 1	ABWR	1350

	Type	MWe net
Lungmen 2	ABWR	1350
Total (2)		2700

also known as Taipei County plant, each unit 1350 MWe gross



Source: World Nuclear Association

Post-Fukushima developments

Following the Fukushima accident in March 2011, the AEC initiated a comprehensive nuclear safety review, and the first phase of this was completed in September. The AEC also strengthened its radiation protection capacity and contingency mechanisms, since

Taiwan is very prone to seismic activity. In January 2012 the AEC said that its post-Fukushima inspections found "no safety concerns" with the six operating nuclear units. It also required Taipower itself to review the nuclear plants' safety margins by following the European Union's reactor stress test requirements.

A review conducted by the European Commission and the European Nuclear Safety Regulators' Group (ENSREG) in 2013 confirmed that the safety standards used by Taiwan's nuclear power plants are generally high and comply with international state-of-the-art practices. However, in the light of earlier AEC stress tests the review recommended that Taiwan should update its assessment of all natural hazards, notably earthquakes and tsunamis, so as to be better prepared for such.

The AEC had planned to invite a follow-up review in three years. However, public opinion of nuclear power changed dramatically following the Fukushima accident and the Democratic Progressive Party elected in January 2016 introduced a policy to phase out nuclear energy.

Subsequent reactor shutdowns

In December 2018 the operating licence of Chinshan 1 expired and Taipower confirmed it was to be decommissioned. The unit had been idle for almost three years following a fuel fault. The fault had been rectified and the unit had been cleared for operation by the regulator, but the government blocked its restart. Unit 2 was shut down in June 2017 due to a transmission failure and was not restarted. It entered the decommissioning stage in July 2019.

The operating licence for Kuosheng 1 was due to expire on 27 December 2021. However, due to a lack of storage in the unit's used fuel pool (see below), the reactor was permanently shut down early, in July 2021.

Following damage to the generator caused by a short circuit, Kuosheng 2 was closed in May 2016 for two years. However, after AEC approval, the reactor was restarted in June 2018. The reactor was permanently shut down on 14 March 2023.

In July 2024 the 40-year operating licence of Maanshan 1 expired and the unit was shut down, in accordance with Taiwan's nuclear phaseout policy. Maanshan 2 was shut down in May 2025.

Just before the final closure of Maanshan 2, in May 2025, Taiwan's legislature approved an amendment proposed by the main

opposition party (Kuomintang) to the Nuclear Reactor Facilities Regulation Act, allowing nuclear reactor lifespans to be extended from 40 to 60 years. The amendment could have potentially delayed the nuclear phase-out by up to two decades but came too late to prevent the unit's shutdown. Later, in a referendum held in August 2025, a majority voted in favour of restarting Maanshan 2, but the turnout was below the threshold for the proposal to pass.

Fuel cycle

All materials and services are imported.

Low-level radioactive waste

A low-level radioactive waste storage facility is operated on Lan-Yu island by Taipower.

Used fuel pools

Storage capacity of used fuel pools was an issue in Taiwan for many years. Policy for used fuel has been direct disposal, though reprocessing has been considered. Maanshan had sufficient storage for used fuel in its pools. However, capacity was an issue at Kuosheng and Chinshan.

Taipower contracted with the AEC's Institute for Nuclear Energy Research (INER) for a dry storage facility at Chinshan using US NAC's Universal MPC technology. An operating licence for the dry storage facility at Chinshan was refused by the New Taipei City government.

Taipower contracted with the INER for a dry storage facility at Kuosheng using NAC Magnastore technology – 27 casks each holding 87 fuel assemblies, total 2349 assemblies. The AEC issued a licence for the Kuosheng dry storage facility in August 2015, when there were 8616 fuel assemblies in almost-full ponds (1449 tonnes), but other approvals for the Kuosheng facility were blocked by New Taipei City government. With nowhere to put used fuel – and therefore the inability to refuel – Taipower shut down Kuosheng 1 in November 2016. In April 2017 the AEC approved Taipower's application to convert the cask loading pool at Kuosheng 1 to store fuel. Dry cask storage had been rejected by the local government.

Reprocessing

In October 2014 a government task force recommended that used fuel from Chinshan and Kuosheng be sent abroad for reprocessing. With MOEA backing, in February 2015 Taipower announced a tender for reprocessing 1200 BWR fuel assemblies from Chinshan (480) and Kuosheng (720) to test the feasibility of this as a general policy. The scope included delivery of transport casks, loading used fuel into them, transport and shipment, reprocessing of the used fuel, management and retransfer of the materials arising from the reprocessing. A contract for the work was expected to cost up to $356 million.

Under the terms of the nuclear cooperation agreement with the USA, the USA agreed to the used fuel being transported overseas for reprocessing, though the agreement specified that all fissionable material would remain with the reprocessor for use in "third party civilian reactors" rather than repatriated. The separated high-level waste would be returned, vitrified, within 20 years for disposal. In March 2015 the tender was suspended pending a parliamentary budget review.

In June 2015 the government set up a committee to decide whether to allocate funds for the project. It did not proceed.

Geological repository

In April 2015, the AEC provided Taipower with guidelines on the technical standards for site selection, based on site selection criteria of the International Atomic Energy Agency, the Nuclear Waste Management Organization of Japan and the US Nuclear Regulatory Commission. The directions required that any final disposal facility be in a bedrock site that is not near densely populated areas, major seismic faults, volcanoes or other changing geological structures, or surface or underground water that could compromise the safety of storage of highly radioactive waste material or risk harm to the geologic environment.

A geological repository in granite for high-level waste is envisaged for 2055 operation. Taipower submitted a report to the AEC in December 2017 summarizing an 11-year study to characterize and evaluate host rock. The second stage of site selection, to be completed by 2028, would identify specific candidate sites. Taipower has been prioritizing granite host rocks based on well-

advanced repositories in Finland and Sweden. It has a technology exchange agreement with Sweden's SKB.

Decommissioning

Taiwanese law requires that applications for decommissioning must be filed by the licensee three years prior to the scheduled final shutdown of a reactor and that it must be approved by authorities before decommissioning can commence.

In January 2016 Taipower published a decommissioning plan for Chinshan. Decommissioning is to be over 25 years, in four stages: shutdown and defuelling to end of 2026; dismantling to 2038; testing to 2041; and site restoration to 2044. The used fuel pool would be removed over 2027-31. Similar plans were published for Kuosheng in 2020 and Maanshan in 2023.

Research & development

There have been six research reactors on Taiwan. Only THOR, a 2 MW Triga unit, at National Tsing Hua university is operating, with the rest shutdown and being decommissioned.

TRR, a 40 MW heavy water research reactor, the largest of the six to have operated on Taiwan, was shut down in 1987. It was to be redesigned as a light water reactor but has been dismantled.

Organization, regulation and safety

The Atomic Energy Council (AEC) consists of representatives from relevant government ministries. The Radwaste Administration is a subsidiary body and is regulator in respect to radioactive waste.

The Nuclear Regulatory Division is also part of the AEC, as is the Radiation Protection Division. The AEC is also responsible for safeguards. The AEC's technical support subsidiary is the Institute of Nuclear Energy Research (INER).

The Atomic Energy Law came in to force in 1968 and various regulations have been promulgated under it.

In 2012 a new Nuclear Safety Authority was to be established to take over from AEC as regulator, and the AEC was to be merged with the Ministry of Science & Technology. The INER would be

moved to the Ministry of Economic Affairs (MOEA) and the 'nuclear' focus dropped. However, in February 2013 cabinet decided to downgrade the AEC from ministerial level and turn it into a safety inspection commission – Nuclear Safety Commission – directly under cabinet. This has apparently not happened, and the AEC continues its regulatory role in respect to nuclear safety.

In 2011 Taiwan and mainland China signed an agreement on nuclear safety and emergency reporting. Under the agreement, China and Taiwan will provide each other with information on their nuclear power plants, regulations and standards for nuclear safety, and exchange their experiences in nuclear safety and plant ageing. It also calls for them to cooperate on nuclear incident communications rated at INES Level 2 or above, environmental radiation monitoring, as well as emergency preparedness and response. Taiwan's AEC says that the agreement will promote cross-Strait nuclear power safety information transparency and enhance the safety and operating performance of nuclear power plants so as to ensure the safety of people on both sides.

The USA had a nuclear cooperation agreement with Taiwan dating from 1972 and running to June 2014. This was renewed indefinitely in December 2013, and entered into force in June 2014 after review by the US Congress. In relation to this, the USA has authorized Taiwan to engage in reprocessing arrangements with France if it later wishes to do so, though “special fissionable materials” derived from the reprocessing, such as plutonium, would remain in France and the remaining waste material would be returned to Taiwan.

Non-proliferation

All nuclear facilities on Taiwan are subject to a non-governmental safeguards agreement with the International Atomic Energy Agency, and all fall under full safeguards.

Taiwan signed the Nuclear Non-Proliferation Treaty (NPT) in 1968 and ratified it, but after 1971, the People's Republic of China replaced Taiwan in the NPT and the IAEA. In terms of such treaties and organizations, and for those countries which adhere to a one-China Policy, Taiwan does not exist as an independent state. The USA recognizes Taiwan as an independent state and has state-to-state relations with it. Taiwan has a unique status. Nuclear safeguards are applied in Taiwan under a trilateral agreement between Taiwan, the USA and the IAEA.

Thus, the IAEA applies safeguards in Taiwan to all nuclear material and nuclear facilities as if it were an NPT non-nuclear-weapon state party; it conducts regular inspections including Additional Protocol verification activities.

Notes & references

Taiwan Atomic Energy Council
Taiwan Bureau of Energy